

Probability Essentials for Deep Learning:

The Big Idea:

- models don't say "this is a cat";
- they say "92% sure it's a cat";
- Probability is how we work with this uncertainty

cat: 0.92 dog: 0.08 → a confidence score; not a fact.

3 Questions Every Neural Network Asks:

How Sure Am I?

Sigmoid / Softmax

How wrong was I?

Cross-Entropy Loss

How do I start?

weight init - Gaussian

1. Outputs to Probabilities:

img → NN → logits (raw scores)

cat → 2.7 → cat : 0.85

dog → 1.3 → dog : 0.15

Sigmoid

1 logit → 1 probability

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

logit = 2.0 ; $\sigma(2.0) = 0.88$

Use: Binary Classification

Softmax

many logits → many probabilities → sum to 1

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

(2.0, 1.0, 0.1) → [0.66, 0.24, 0.10]

Use: Multi-class classification

2. Cross - Entropy Loss :

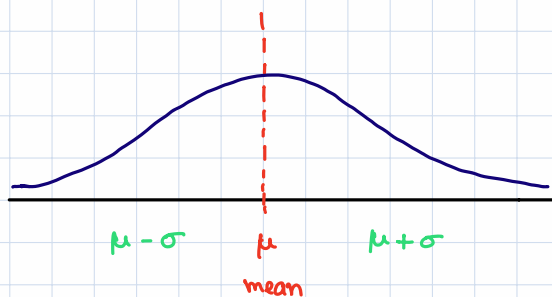
- compares pred. probs vs true labels
- Punishes **confident wrong** answers the **hardest**

$$L = - \sum y \cdot \log(\hat{y})$$

$$\begin{aligned} &\geq 0.5 \rightarrow 1 \\ &< 0.5 \rightarrow 0 \end{aligned}$$

True Labels	Pred Labels	Penalty
cat (1)	0.85 $\rightarrow$ 1	Low
cat (1)	0.45 $\rightarrow$ 0	Medium
cat (1)	0.05 $\rightarrow$ 0	High

3. Guassian Distribution (Bell Curve)



μ (mean) $\rightarrow$ center

σ (std. dev) $\rightarrow$ spread

- a way to generate random numbers.
- most values fall around the center, and fewer at the edges
- used to initialize weights at small random values
- Batch Normalization

The 3 Math Pillars of DL:

Linear Algebra

$\rightarrow$ store and move the data

Calculus

$\rightarrow$ tells the model how to learn

Probability

$\rightarrow$ model's way of saying how sure it is